
HCI 520 P3 E-Learning Evaluation Project

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Objectives

Honestly, for this project, I really want to move beyond just making design decisions because they "look right" or fit my gut feeling. After doing the first two assignments, I've realized that I used to rely a lot on intuition, but now I'm trying to be way more intentional. My main goal here is to see if professional systems actually follow the e-learning guidelines we're studying, like the Multimedia and Contiguity principles, and how those design choices move a learner from just memorizing facts to actually understanding a process. I'm especially interested in seeing how they handle feedback, since I've been thinking a lot about how visual feedback helps people master lighting.

Method

To get this evaluation done, I took a "deep dive" into two different worlds. First, I spent about 50 minutes on Chomp Chomp (Grammar Bytes), specifically going through the "Misplaced and Dangling Modifiers" section. I acted like a student who was totally lost to see how the system handles mistakes. Then, I put my own Tabletop Photography system under the microscope. I compared my wireframes and the "Virtual Shoot" module against the 12 principles from the ELSI book to see where I hit the mark and where I might have made things too complicated for a beginner.

Results

System Name:	Chomp Chomp (Grammar Bytes)		
Intended Audience:	Students who need to document their work or fix their writing style.		
Learning Objectives (Bloom-style)	Recognize misplaced modifiers and apply correction rules (Bloom: 1.1, 3.1).		
Principle	Evaluation (Followed / Violated / Partial / NA)	How? (Briefly describe)	Impact on learning (describe / speculate)

Multimedia	Followed	Uses goofy little cartoons that show the sentence literally acting out the mistake.	Makes the grammar rule easier to remember because the joke sticks in your head.
Contiguity	Followed	The correction shows up right under the question you got wrong instead of somewhere else.	You immediately connect the mistake with the explanation without getting lost.
Modality	Violated	Everything is just reading and clicking with no narration or audio support.	Can get tiring because learners are constantly reading both the problem and explanation.
Redundancy	N/A	Since there's no narration, there's no issue with repeated audio and text.	Doesn't really apply here.
Coherence	Followed	The site keeps things simple and doesn't overload the page with random extras.	Helps learners stay focused on the grammar instead of distractions.
Personalization	Followed	The site talks to the learner casually instead of sounding like a strict teacher.	Makes grammar feel less intimidating and more approachable.
Segmenting and Pretraining	Followed	Every activity is short and broken into tiny practice chunks.	Prevents learners from feeling overwhelmed all at once.
Worked Examples	Partial	Gives some examples before practice, but I think it could walk through them more clearly step-by-step.	More guided examples would help learners understand the "why" before trying alone.

Practice	Followed	Most of the site is interactive grammar drills and corrections.	Repetition helps learners actually apply the rules instead of memorizing definitions.
Collaborative	Violated	There's no feature for working with classmates or comparing answers.	Misses out on peer discussion and social learning opportunities.
Learner Control	Followed	Learners can pick whatever grammar topic they want and move at their own speed.	Gives users more control and makes the experience feel less forced.
Thinking (general problem solving) skills	Partial	Mostly focuses on finding grammar mistakes instead of improving overall writing style.	Good for accuracy, but not as strong for creative or higher-level writing skills.
Feedback	Followed	When you miss something, it explains exactly what part of the sentence went wrong.	Helps learners understand their mistakes instead of randomly guessing again.
Far Transfer	Partial	The activities feel more like quizzes than real-world writing situations.	Learners might still struggle using the skill naturally in essays or emails.
Deep understanding	Partial	The focus is mainly on spotting errors instead of creating original writing.	Builds foundational understanding, but not deeper mastery.
Other(Humor)	Followed	Uses cheesy jokes and dramatic wrong-answer sounds that make the site feel playful.	Makes grammar practice less stressful and honestly more fun.

System Name:	Tabletop Photography (My Project)		
Intended Audience:	Design students or makers who need to photograph models or prototypes.		
Learning Objectives (Bloom-style)	Identify 3-point lighting components and execute a virtual lighting setup.		
Principle	Evaluation (Followed / Violated / Partial / NA)	How? (Briefly describe)	Impact on learning (describe / speculate)
Multimedia	Followed	Shows both the lighting setup and the final photo render at the same time.	Helps learners connect what they change in the setup to what happens in the final image.
Contiguity	Followed	The lighting controls, setup view, and photo result are all side-by-side.	Makes it easier to understand cause and effect without getting confused.
Modality	Partial	The system mostly relies on visuals and text with little narration right now.	Could become tiring during longer sessions because learners are reading a lot.
Redundancy	Followed	Avoids repeating the same information through both narration and text.	Keeps the experience cleaner and prevents unnecessary overload.
Coherence	Followed	Removes extra photography topics like ISO or shutter speed to focus only on lighting.	Helps learners stay focused on mastering lighting first.
Personalization	Followed	Uses a casual and creative tone instead of sounding overly technical.	Makes the experience feel more approachable for design students.
Segmenting and Pretraining	Followed	Breaks the lesson into smaller sections before the final virtual shoot activity.	Prevents learners from feeling overwhelmed too early.

Worked Examples	Followed	Includes a guided lighting demo before learners try their own setup.	Gives learners a clear example to follow before experimenting independently.
Practice	Followed	Learners can drag lights around and test different placements themselves.	Encourages hands-on learning and experimentation.
Collaborative	Followed	Includes peer critiques where students discuss and explain lighting choices.	Helps learners build communication and evaluation skills.
Learner Control	Followed	Users can freely move and adjust lights in the virtual studio.	Encourages exploration and creative decision-making.
Thinking (general problem solving) skills	Followed	Learners solve problems like reducing harsh shadows while still keeping depth.	Builds critical thinking and creative problem-solving skills.
Feedback	Followed	The image updates instantly whenever the learner changes the lighting.	Immediate feedback helps learners quickly understand the effects of their choices.
Far Transfer	Partial	The project focuses mostly on virtual setups rather than real camera equipment.	Learners may still need help applying the skills in real-world photography situations.
Deep understanding	Followed	Learners move from identifying lighting setups to creating and critiquing their own.	Encourages higher-level understanding and creative application.
Other (Visual Realism)	Followed	The studio and rendered images are designed to look realistic and professional.	Motivates learners by making the activity feel authentic and portfolio ready.

Discussion

One thing I found really interesting was how much tone and presentation can completely change the learning experience. Chomp Chomp teaches something that most students usually find boring, grammar, but the humor, cartoons, and casual writing style make it feel way less stressful. Even though it is simple, the design choices make the activities more memorable.

For my Tabletop Photography project, I noticed how important immediate visual feedback is for creative learning. Instead of just reading about lighting, learners can actually move lights around and instantly see how shadows and depth change in the final image. That kind of interaction feels much more engaging than passive instruction.

Another interesting point was that both systems focus heavily on practice and feedback, but in different ways. Chomp Chomp focuses on correcting mistakes directly, while my project focuses more on experimentation and exploration.

Reflection questions

Which of the systems do you think are most likely to reach their learning goals and why? Which systems are most likely to fail?

I think both systems are likely to reach their core learning goals because they focus strongly on practice and immediate feedback. Chomp Chomp is effective for helping learners recognize grammar mistakes because it gives instant explanations and keeps the activities short and focused. My Tabletop Photography project is likely to succeed because learners can directly experiment with lighting setups and immediately see the results of their choices.

However, both systems also have limitations. Chomp Chomp may struggle with deeper learning because learners are mostly identifying errors instead of applying grammar naturally in real writing situations. My project could struggle with far transfer because learners might understand lighting virtually but still have difficulty setting up real equipment in an actual studio environment.

Did you see examples in the systems where the guidelines were not followed, but where you thought it was justified? Explain.

Yes. In Chomp Chomp, the modality principle is technically violated because there is no narration or audio support, but I think it makes sense for the type of system it is. Adding narration for every grammar question might actually slow learners down or become repetitive. Since the exercises are short and simple, text-only interaction still works reasonably well.

In my photography project, I intentionally left out extra photography concepts like ISO and shutter speed, even though some learners might expect them. I think this is justified because the goal is specifically to teach lighting. Including too many camera settings at once would probably overwhelm beginners and distract from the main learning objective.

If you were to categorize the systems that you examined, how would you do it? What types of categories were there, and how did the systems fit in? (You can use whatever existing categories, or invent new ones based on your experience.)

I would probably divide the systems into skill-drill systems and exploratory creative systems. Chomp Chomp fits into the skill-drill category because it focuses on repetition, correction, and mastering a specific rule-based skill. The learner practices until they recognize patterns automatically.

My Tabletop Photography project fits more into an exploratory creative learning category because learners are experimenting, making design decisions, and analyzing visual outcomes. There is not always one “correct” answer, which makes the experience more open-ended and creative.

I also think another category could be corrective learning versus discovery learning. Chomp Chomp mainly corrects mistakes, while the photography project encourages learners to discover what works through experimentation.

What do you think is the relationship between the learning objectives for a system (e.g. levels of cognitive processes in Bloom's taxonomy) and the types of design elements it uses?

I think the learning objectives heavily influence the design choices in a system. Lower-level Bloom's objectives like remembering or recognizing usually work well with quizzes, repetition, and direct feedback. That is why Chomp Chomp focuses so much on short practice exercises and corrections.

Higher-level objectives like analyzing, evaluating, or creating need more interactive and open-ended design features. In my photography project, learners are experimenting with lighting setups, predicting outcomes, and critiquing designs, which requires tools like real-time feedback, learner control, and visual simulations.

Basically, the more advanced the learning goal becomes, the more the system needs to support exploration, creativity, and decision-making instead of just memorization.

Which design features of the systems are you most likely to include or avoid in your system design?

I would definitely include immediate feedback because both systems showed how important it is for keeping learners engaged and helping them understand mistakes quickly. I also liked the use of learner control, especially in the photography project where users can experiment freely without being forced through one strict path.

I would also include personalization and conversational tone because it makes the system feel less intimidating and more human. Chomp Chomp does this really well with humor and casual language.

One thing I would avoid is overwhelming learners with too much information at once. I think coherence and segmenting are really important, especially in creative subjects where there are already many variables to think about. I would also avoid relying too heavily on repetitive drills without real-world application because learners may struggle with transfer outside the system.

Conclusions and "Take home lessons"

Overall, this evaluation showed me that successful e-learning systems are not only about providing information, but also about how that information is presented and practiced. Features like immediate feedback, learner control, humor, and interactive practice can make a huge difference in engagement and understanding.

I also learned that different learning goals require different types of design. Systems focused on memorization or recognition benefit from repetition and correction, while systems focused on creativity or problem-solving need exploration, experimentation, and open-ended interaction.

If I were designing future e-learning systems, I would focus on keeping lessons segmented, visually clear, and interactive while still giving learners freedom to experiment. I would also try to improve far transfer by connecting activities more directly to real-world situations. If I repeated this evaluation again, I would probably compare more systems across different subjects to better understand how design changes depending on the type of learning being supported.